CS403

Final Term Prepration

Lecture no 19:

Functional Dependency

- A functional dependency is a type of relationship between attributes.

- Attributes of set of attributes on the left side are called determinant.

- $A \longrightarrow B$: A determines B
- If A and B are attributes or set of attributes of relation R, we say that B is functionally dependent on A if each value of A in R has associated with it exactly one value of B in R.

→ F.D and Keys :

- Super key is an attribute or set of attributes that identifies an entity uniquely.

- FDs helps in finding out their keys and their relation as well.

Inference Rules

Rules of Inference for functional dependencies, called inference axioms or Armstrong axioms.

Different inference Rules :-

1- Reflexivity :

If B is a subset of A, then $A \rightarrow B$.
This is also implies that $A \rightarrow A$ always holds.

2- Augmentation:

If we have $A \rightarrow B$ then $AC \rightarrow BC$.

3- Transitivity:

If $A \rightarrow B$ and $B \rightarrow C$, then $A \rightarrow C$.

4- Aditivity or Union:

If $A \rightarrow B$ and $A \rightarrow C$, then $A \rightarrow BC$.

5- Projectivity or Decomposition:

If $A \rightarrow BC$, then $A \rightarrow B$ and $A \rightarrow C$.

6 Pseudo transitivity:

If $A \rightarrow B$ and $CB \rightarrow D$, then $AC \rightarrow D$.

Normal Form

There are two goals of the normalization process:

1- Eliminate redundant data

2- Ensure data dependencies make sense

First Normal Form

A table is in 1NF if:

1- Each cell contain a single value.

2- Each column has a unique name.

3- Each row is unique.

The table has no repeating columns.

Second Normal Form

A table is in 2nd Normal form if it is:

1- It is in 1st NF.

2- All non-key attributes are full functionally dependent on the key.

(There should be no partial dependency of the attributes.)

Lecture No 20:

2NF

If relation is not in 2NF then there are some "anomalies":

- Redundancy
- Insertion Anomaly
- Deletion Anomaly
- Updation Anomaly

Third Normal Form

A table is in 3rd Normal Form if:

1- It is in 2nd NF.

2- Every non-key attribute is non-transitively dependent upon its primary key.

Boyce-Codd Normal Form:

A table is in BCNF if:

1- It is in 3rd NF.

2- There are no transitive dependencies.

3- For every determinant, it must be a candidate key (ie, it can uniquely identify each row).

Ensure that each table has a single value, unique reason for existance and that each attribute depends on the entire Primary key.

Date: CS 403

Lecture No 22 :

→ Physical Database Design Considerations & Implementation :

Some steps should have discussed before moving onto the Physical Db design :

1) Normalization of relations
2) Volume estimate
3) Definition of each attribute/field
4) Description of where & when data is used (with f)
5) Description of the technologies

→ Designing fields :

- Smallest unit of application.
- Df means what's the name of attribute / field, size and data type of that field etc.

* Data type defined

4 major objectives for using data types :

1) Minimized usage of storage space
2) Represent all possible values
3) Improve data Integrity
4) Support all data manipulation

Date:

→ Coding & Compression Techniques:

Coding techniques are useful for compression of data values appearing the data, by replacing those data values with the smaller sized codes.

Reading ⟶ R (code)

Gardening ⟶ G "

→ Benefits by the use of datatypes:

1) Default Values
2) Range Control
3) Null Value Control
4) Referential Integrity

"Logical connection b/w two tables with the help of Primary key as a foregin key".

* Pk & Fk both have the same data type. It's necessary for connection.

Date:

Lecture No 23:

→ Denormalization:

This process is applied for deriving a physical data model from a logical form.

* To move from "higher to lower" normal forms. Due to "speed up" of database access.
* Leads to merging different relation.

→ Partitioning:

* Partitioning splits same relation into two.

→ Aim / Purpose:

1) Reduced workload (e.g; data access)
2) Balance workload
3) Speed up the rate of useful work

→ Types

There are two types of Partitioning.

1) Horizontal Partitioning
2) Vertical Partitionig

Date:

1) Horizontal Partitioning:

Table is split on the basis of rows, which means larger table split into smaller tables. Some of types of HP are as under:-

- **Range Partitioning:**

Partitioning range is imposed on any particular attribute.

- **Hash Partitioning:**

Hash Partitioning reduces the chances of unbalanced partitions to a large extent by particular algorithm. DBMS knows that algorithm.

- **List Partitioning:**

Partitioning the values are specified for every partition. So, there is a specified list for all the partitions.

* There is no range involved in this.
* There is a list of values.

Date:

Lecture No 24:

2) Vertical Partitioning:

Table is split on the basis of fields.
Same table is split into different physical records depending on the nature of access.

* Primary key is repeated in all vertical partitions of a table to get the original table.

→ Replication:

The process of copying a portion of database from one environment to another and keeping subsequent copies of the data in synchronization with the original source.

↓

Changes made to the original source are propagated to the copies of data in other environments.

* Replication is the final form of denormalization.

* It increases the access speed.

* It decreases failure damage of the database.

Date:

* Replication is normally adopted in those applications, where updation is not very frequent because if updation is frequent so then it will have problems of updation in all the copies of database relations.

→ Clustring Files:

The process which means to place records from different tables to place in adjacent physical locations, called clusters.

* Cluster increase the efficiency.

Mcqs ⇒ Data Definition Language (DDL) skills to translate physical design into actual database objects.

→ Transforming Logical into Physical data model:

1) Transforming entities into tables
2) Transforming attributes into columns
3) Transforming domains into data types and constraints.

Date:

There are many decisions that must be made during the transition from Logical to Physical.

For example, each of the following must be addressed:

1) The null ability of each column in each table.
2) For character columns, should fixed length or variable length be used.
3) DBMS used to assign value to sequences.
4) Implementing logical relationships by assigning ~~reff~~ referential constraints
5) Building indexes on columns to improve query performance
6) Deciding on the clustering sequence for the data.
7) Other physical aspects such as column ordering, buffer pool specification, data files, denormalization, and so on.

Date:

→ Structured Query Language

- Any type of interaction with database through DBMS.
- SQL is an ANSI standard computer language for accessing and manipulating databases.
- SQL is standardized, and the current version is ~~reffr~~ referred to as SQL-92.
- SQL is used to communicate with a database.

* RDBMS that use SQL are :
Oracle, Sybase, Microsoft SQL Server, Access, Ingres, etc.

→ Standard SQL commands :

"Select", "Insert", "Update", "Delete", "Create" and "Drop".

→ Benefits of Standard SQL :

1) Reduced training cost
   - Application Portability
   - Application longevity
   - Reduced dependence on a single Vendor
   - Cross-System Communication

Date:

Lecture No 25:

→ SubLanguages of SQL Commands:

SQL commands can be divided into two main sub languages.

1. Data Definition Language (DDL)
2. Data Manipulation Language (DML)

→ Rules of SQL Format:

- Reserved words are written in capital like SELECT or INSERT.
- User-defined identifiers are written in Lower case.
- Identifiers should be valid. The maximum length can be 256.
- The reserved words should not be use as identifiers.
- Those things in the command which are optional are kept in [ ].
- Curly braces { } means required items.
- | means choices
- [,.....n] means n items separated by Comma.

Date:

1) DDL

The **data dictionary** contains meta data (data about data) the storage structure and access methods used by the database system are specified by a set of definitions in a special type of DDL called a data storage and definition language.

DDL statement are compiled, resulting in a set of tables stored in a special file called a **data dictionary** or **data directory**.

2) DML

A DML is a language, which enables users to access and manipulate data. Data Manipulation is retrival, insertion, deletion, modification of information from the database.

Types

1- **Procedural:** in which the user specifie What data is needed & how to get it.

2- **Nonprocedural:** in which the user specific Only what data is needed.

Date:

Lecture no 26:

→ Categories of SQL Commands

Three different types of commands of SQL, which are DDL, DML and DCL.

1) DDL:

It deals with the structure of database. The DDL (Data Definition Language) allows specification of not only a set of relations, but also the following for each relation:

- The schema for each relation.
- Domain of values associated with each attribute.
- Integrity constraints.
- The set of indexes for each relation.
- Security and authorization information.
- Physical storage structure on disk.

→ Three different commands of DDL :-

1) Create
   - Through SQL Create command
   - Through Enterprise Manager
2) Alter
3) Truncate

default value of the

Date:

Exercise: Type of DDL command

Create a database of Exam System and create a table of student with different Constraints in SQL Server.

```
-- CREATE DATABASE
CREATE DATABASE Exam;

-- Create Table Name Student without
  Constraints
CREATE TABLE Student;
(
   stdID    INT ,
   stdName  TEXT ,
   CGPA     REAL ,
   Semester SMALLINT
);
-- Create Table Name Student with constraints.
  CREATE TABLE Student 1;
(    stdID    INT PRIMARY KEY,
     stdName  TEXT NOT NULL,
     CGPA     REAL CHCK (CGPA < 4.00 OR CGPA > 0.00),
     Semester SMALLINT DEFAULT 1
);
```

Date:

Lecture No 27:

2) Alter Table statement

Purpose:

ALTER statement is to make changes in the definition of a table already created through Create statement. It can add, drop of the attributes or constraints, activate or deactivate constraints. It modifies the design of an existing table.

Syntax:

```
ALTER TABLE table {
ADD [COLUMN] column type [(size)] [DEFAULT default] |
ALTER [COLUMN] column type [(size)] [DEFAULT default] |
ALTER [COLUMN] column SET DEFAULT default |
DROP [COLUMN] column |
RENAME [COLUMN] column TO columnNew
}
```

→ Using ALTER TABLE

- Use ADD COLUMN to add a new column to the table. Specify the name, data type, size & default value of the column.
- Use ALTER COLUMN to alter type, size or default value of an existing column.

Date:

- Use DROP COLUMN to delete a column.
- Use RENEME COLUMN to rename an existing column.

We can't add, delete or modify more than one column at a time.

3) Truncate Command

Purpose:

The TRUNCATE is used to delete all the rows of any table but rows would exist. If we want to remove all records we must use TRUNCATE Command.

Syntax:

```
TRUNCATE TABLE table_name
```

→ Delete

The DELETE is use to delete one or many records.

→ DROP

The DROP table command, which is used to drop the comple table from database.

Syntax:

```
DROP TABLE table_name
```

Date:

2) DML

The Data Manipulation Language is used to retrive, insert, delete and modify database information.

→ Three different Command:

1) Insert ----> (add new rows)
2) Select ----> (retrive rows)
3) Update ----> (modif the rows)

1) Insert Statement

The insert command in SQL is used to add records to an existing table.

It has two Formats:

INSERT INTO table-1 [(column-list)]
VALUES (value-list)

OR

INSERT INTO table-1 [(column-list)] (specific)
VALUES (value-1 [, value-2]..)

Example:

INSERT INTO Course VALUES ("CS-211", 'OS', 4, 'BCS')

i.e

Course (crCode, crName, crCredits, prName)

Date: Lec No 28:

2) Select Statement

It is used to select rows and columns (data) from table in database.
The basic SELECT statement has 3 clauses:

- SELECT ----> specifies the column
- FROM ----> specifies the table
- WHERE ----> specifies which table rows are used.

Queries

Q1) If you want to select all the columns.

```
SELECT * Student ;
       table name
```

Q2) If you want to select specific columns.

```
SELECT col1, col2 FROM tablename ;
```

Q3) If you want only unique items selected from column. (ignoring repeated)

```
SELECT DISTINCT col_name FROM tablename;
```

4) If you want to show a New Name of column that you set.

```
SELECT col_name AS 'ANY NAME' FROM tablename;
```

Date:

Lecture No 29:

Q : Display all courses of the MCS Program

```
SELECT CrName, PrName FROM tablename;
WHERE PrName = 'MCS';
```

Q : List the course names offered to Programs other than MCS from course table

```
SELECT crcode, crName, prName FROM Course
WHERE PrName != MCS ;
-- WHERE NOT (PrName = 'MCS');
```

} Same result

Q : Display the names and credits of CS programs

```
SELECT CrName, CrCredits, PrName FROM Course
WHERE PrName LIKE '%CS';
```

Date:

Lecture No 30:

→ ORDER BY Clause

The ORDER BY clause allows you to sort the records in your result set. The ORDER BY clause can only be used in SELECT statements. The result set based on the columns specified. After column name write the order ASC or DESC

```
SELECT * FROM tablename (Student)
WHERE stId > 100
ORDER BY stId ASC;
```

→ Function in SQL

A function is a special type of command. Functions are one-word command that return a single value. e.g; CURRENT_TIME

Built-in:- which are provided by any specific tool or language.

User-defined:- which are defined by the user.

Date:

→ Categories of Functions:

These categories of functions are specific to SQL Server. Functions are categorized as under:

- Mathematical (ABS, ROUND, SIN, SQRT)
- String (LOWER, UPPER, SUBSTRING, LEN)
- Date (DATEDIFF, DATEPART, GETDATE())
- System (USER, DATELENGTH, HOST_NAME)
- Conversion (CAST, CONVERT)

Example:-

```
SELECT UPPER(stName), LOWER(stFName), stAdress
LEN(CONVERT(char, stAdress)) FROM student;
```

→ Aggregate Functions

These functions operate on a set of rows and return a single value.

If used among many other expressions in the item list of a SELECT statement, then must have a GROUP BY clause.

No GROUPBY clause is required if the aggregate function is the only value retrived by the SELECT statement.

Some Aggregate Functions :- - - - -

Date:

Aggregate Functions

Function	Usage
• AVG (expression)	---→ Computes average value of a column by the expression.
• COUNT (expression)	---→ Counts the rows defined by the expression.
• COUNT (*)	---→ Counts all rows in the specified table or view.
• MIN (expression)	----→ Find the minimum value in a column by the expression
• MAX (expression)	----→ Find the maximum value
• SUM (expression)	---→ Computes the sum of col value

eg; SELECT COUNT (marks) FROM tablename;

Date:

→ GROUP BY Clause

Used in a SELECT statement to collect data across multiple records and group the results by one or more columns.

Example by using SUM function:

```
SELECT departement, SUM (sales) AS 'Total Sales'
FROM Order-details
GROUP BY departement;
```

→ Having Clause

The HAVING clause is used in combination with the GROUP BY clause. It is used to filter the records that a GROUPBY returns.

```
SELECT AVG (marks) FROM student
WHERE stID > 100
GROUP BY marks HAVING marks > 300;
```

→ Cartesian Product

A cartesian join gives a cartesian product.
A cartesian join is when you join every row of one table to every row of another table.
You can also get one by joining every row of a table to every row of itself.

Date:

Lecture No 31:
→ Types of Joins

1- Inner Join:

The inner join "between two tables" can be performed "on the basis" of the "common attribute" between the two. The common attributes are not required to have the same name in both tables, however, they must have the same domain in both tables.

I have two tables:

1) Course (crCode, crName, PrName)
2) Program (PrName, TOTsemester, Pr Credits)

SQL query:

```
SELECT * FROM Course INNER JOIN Program
ON Course.PrName = Program.PrName ;
```

2- Outer Join:

Join operation that rely on null values, called outer joins.

Two types of Join:

1) Left outer Join
2) Right Outer Join

Date:

SQL query for Outer Join:

```
SELECT * FROM Course OUTER JOIN Program
ON Course.PrName = Program.PrName;
```

1) Left outer Join

All the attributes values display of the Left table but only common attribute will be display of the right table.

```
SELECT * FROM Course LEFT OUTER JOIN
Program ON Course.PrName = Program.PrName;
```

2) Right outer Join

All the values of right table display but from the Left table, only common values display.

```
SELECT * FROM Course RIGHT OUTER JOIN
Program ON Course.PrName = Program.PrName;
```

3- Semi Join:

First inner join is performed on the participating tables and then resulting table is projected on the attributes of one table. We can select specific rows of table 1st that have matching the rows in 2nd table.

Date:

SQL query for Semi Join:

```
SELECT Course.crCode, Program.prName,
Program.prCredits FROM Course SEMI JOIN Program
ON Course.prName = Program.prName;
```

4- Self Join:

In self join a table is joined with itself. All the attributes displayed 2 times of a single table.

```
SELECT * FROM Course AS c1 , Course AS c2
WHERE c1.crCode = c2.crCode;
```

→ Subquery

It is also called nested query. It is one of the most powerful features of SQL. A query In which another query embedded ; the embedded query is called subquery".

Example:

```
SELECT * FROM Student
WHERE marks > (SELECT marks FROM Course
WHERE course code > 100 );
```

Txte

3) → DCL

Data Control Language:

1. GRANT
2. REVOKE

SQL-92 supports access control through the GRANT and REVOKE commands.

1. GRANT

A DCL command that allows Providing privileges for the users on the database objects.

- Allows assigning access rights to the users.
- Grant is easier to execute than Rewoke.

SQL query:

GRANT SELECT, INSERT ON databaseName.tableName TO User WITH GRAT OPTION ;

2. REMOKE

A DCL command that allows taking back the permission assigned to a user.

- Allows removing the access rights from the users.
- If access is removed from one user, all other user will also lose access.

REVOKE SELECT, INSERT ON database.Name.tableName FROM user CASCADE (Executed BY host/saved) ;

Lecture No 32 :

→ Application Programs

Application programs can be developed in parallel or after the construction of database design. Tool selection is depend upon developer in which he feels comfortable

General Activities in Application Programs :

- Data input Program
- Editing
- Display
- Processing related to activities
- Reports

→ User Interface :

A window or screen use for Communication between User & System or Computer.

→ Types

1) Text based
2) Graphical User Interface ---> most commonly called as Forms.

1) Text based User Interface

In this interface certain keyboard numbers are designated for any action.

For example:

Adding a Record	1
Deleting a Record	2
Enrollment	3
Result calculation	4
Exit	5

2) Forms

Forms are now days used extensively in the application programs.

Following are the different types of forms:

1) Browser Based:

These are web-based forms. They are developed in HTML, scripting language or Front Page.

2) Non-Browser Based / Simple.

→ **User Friendly Interface:**

- Beginners User
- Intermediates User
- Expert User

Lecture No 33:

Imp For MCQs

Method to create Forms:

1) With Wizard
2) Without Wizard

How many View of Forms in MS Access?

1) Layout View
2) Design View
3) Datasheet View

Microsoft Access = MSAccess

How many View of Table in MS Access?

1) datasheet view
2) design view

Lecture No 34:

→ Classification of Physical

Storage:

According to following characteristics:

speed of access

Cost per unit of data

Reliability

→ Type of Storage

1- Volatile storage:

Computer storage that is lost when the power is turned off. e.g: RAM, cashe etc.

2- Non-Volatile:

Computer storage that is not lost when the power is turned off. e.g; Hard disk, USB, etc.

cashe is a special high-speed storage machenism.

1- Memory caching

2- Disk caching

→ Imp MCQs:

Internal caches are often called Level 1 caches

External caches memory, called Level 2 caches.

→ When data is found in the cache, is called a cache hit.

→ The CPU can directly access RAM.
→ Random Access Memory. (RAM)
→ Flash memory is non-volatile.
→ MOS stans for Metal Oxide Semiconductor.
→ Hard Disk can store anywhere from 20MB to more than 10 GB. Hard Disk are also from 10 to 100 times faster than floppy disks.

CD-ROM: You can read the data from a CD-ROM, but you cannot modify, delete, or write new data.

~~WROM~~: → **WORM**

Stands for Write-once, read-many.

→ **Erasable Optical (EO)**

EO disk. can be read to, written to, and erased just like magnetic disks.

→ RAID

Stands for Redundant Array Independent Disk

RAID-0:

Data store only **one time**. RAID level 0 is not redundant. **but** the faliure of any disk in the information is stored array result in data loss.

RAID-1:

Same data stores **many times**.

RAID-1 provide redundancy by writting all the data to two or more drives.

but if either drive fails, no data is lost.

RAID-2:

RAID Level 2, which uses Hamming error correction codes.

disk 1 in data, disk 2 in Parity bit of disk 1 data disk 3 in data, disk 4 in Parity bit of disk 3 data.

RAID-3:

Data store in Multiple disk & Parity bits of all data store in one disk.

- Similar to RAID-4 but for **Large records**.

~~RAID~~

RAID-4:
Similar to level 3 but for small records accessing.

RAID-5:
Parity stores on multiple drives.
Used for Multi Users access.

→ Sequential File Organization

Records are arranged on storage devices in some sequence based on the value of some field, called sequence field.
Sequence field is often the key field that identifies the record.
All the records have the same size and the same field format

→ File Protection

When multiple user have access to files, it may be desirable to control by whom and in what ways file may be accessed. This is called file protection.
Following are the two types:

1. Indexed Sequential
2. Direct File Organization

Lecture No 35:

1-Indexed Sequential file:

An index file can be used to effectively overcome the above-mentioned problem, and to speed up the key search as well.

A type of file access in which an index is used to obtain the address of block which contains the required record.

→ Types of Indexes:

- Exhaustive type
- Partial type

→ Defining key:

* An index file must have at least one key.
* The first (Pk) key is always numbered 0.
* Have up to 255 keys.

Insert a new record or update is directly related to the number of keys.
Retrival time is not directly related to the number of keys.

Indexed sequential file, you must define each key in the following terms:

- Position & size
- Index number
- Data type
- Options selected

In indexed sequential files following are **ensured**:

- New records are added to an overflow file.
- Record 6 point to new record (7).
- Overflow is merged with the main file during a batch update.
- Multiple indexes for the same key field can be set up to increase efficiency.

Lecture No 36:

→ Hashing

A key record field is used to calculate the record address by subjecting it to some calculation; a process called hashing.

- Hashing provides rapid, non-sequential, direct access to records.

Hash Function:

value % 10

Records:

10, 23, 26, 45, 20

Now,

we can find address of the abaved record.

Address	Records
0	10 , 20
3	23
6	26
5	45

Value % 10

10 % 10 = 0
23 % 10 = 3
26 % 10 = 6
45 % 10 = 5
20 % 10 = 0

→ Hash Function

A good hash function is a small constant, independent of the number of search keys.

- The worst hash function maps all keys to the same bucket.
- The best hash function maps all keys to distinct addresses.
- Distribution of keys to addresses is uniform and random.
- Insertion and deletion are simple.

→ Hashed Access Characteristics

- No indexes to search or maintain
- Very fast direct access
- Inefficient sequential access
- Use when direct access is needed, but sequential is not.

Handling the Collisions

Various techniques are used to manage this problem:

1 - Chaining (Closed Addressing / open hashing)

2 - Overflow areas

3 - Re-hashing

4 - Using neighboring slots (linear probing)

5 - Quadratic probing

6 - Random probing

Indexes

Primary key valy column pr index بنتی
create hojaty hain.

Classification

1- Clustered vs Un-Clustered Indexes
2- Single key vs. Composite Indexes
3- Tree-based, inverted files, Pointers

Index must identify a record in table, uniquely.

Types

1- Primary Indexes	2- Secondary Index
• having unique field e.g. Student ID	• Not having unique field. e.g. Student Name • Record sorted on the basis of PK, name, ~~sequential search~~

Three Implementation approaches of Indexes:

1- Inverted files or inversions
2- Linked List
3- B+ Tree

Query for Creating Index

```
CREATE INDEX index-name ON table-name (colum-name);
```

```
CREATE INDEX myIndex ON Student (stdName);
```

Properties of Index

1- Index can be define when there is no data in table.
2- Existing values are checked on execution of this command.
3- It support selections of form (field operator > constant)
4- It support equality selections ("tree" or "hash")
5- It support Range selections (<, >, <=, >=, BETWEEN)

Lecth 38

A A table can contain only one clustered index. • Faster • less memory for operation are efficient for finding a specific row when the index value is unique. • Index is the main data. • has inherent ability to store data on disk.

Non-Clustered: • Slower

- Not stored in order / Not sorted
- Leaf nodes contain index rows.
- Leaf layer doesn't consist of data page.
- More memory for operation required
- In non-clustered index, index is the copy of data
- A table have multiple non-clustered index
- ~~Has not inherent ability to store~~
- Doesn't have inherent ability to stored data.

Type of order indecies

Dense Index:
An index record appears for every searchkey value in file.

Sparse Index:
Index records are created only for some record.

↳ Two-level sparce Index

- Index Update
- Deletion - (key value del, then data del)
- Insertion

Dense: insert search key value if not present

Sparce: no change unless new block is created

39 & 40

Virtual Table

We can customize the order of table through fields

views → **Views**

Views is a virtual table of original base table whose content defined by a SQL Query.

- Used for Security mechanisms. (At small level)
- Grant & Revork used (for High level security)

Two ways to create a new view

1- Create a new view from scratch.

2- Make a copy of an existing view & then modify it.

Types of views.

1. • Materialized View : Permanently stored Physical than base table view
   This view is a pre-computed data set derived from a query specification & stored for later use. • Faster

2. • Simple View: Not stored Physical
   1- Create from only one table (base table).
   2- Can't use group function.
   3- Doesn't contain group of data
   4- DML operations could be performed.
   5- Doesn't contain GroupBy, Distinct, Pseudocolumn

3. • Complex view / Dynamic view : Not stored Physical
   1- Created from more than once base table.
   2- Can use aggregation / group function.
   3- It can contain group of data.
   4- DML operation could not always performed through complex view.
   5- It contains GroupBy, distinct, pseudocolumn
   6- We can't apply Insert, Del, Update on complex view directly.

```
CREATE View view1 AS( SELECT columns_name FROM tablename );

CREATE View view2 AS( SELECT * FROM tablename WHERE condition );

SELECT * FROM view1;
SELECT * FROM view2;
```

41

Materialized View

A Pre-computed table comprising and aggregated or joined data from fact & possibly dimension tables. Also knowns as **summary** or aggregate table.

- Views are Virtual.
- Materialized view also called Indexed view.
- Created through Clustered index.

Applications

1- Data Warehousing:

In data warehouse, materialized views are used to **compute** & **stored** aggregate data such as sum & average.

2- Distributed:

- Used to replicate data at distributed sites & synchronize updates

3- Mobile Computing:

- Used to download a subset of data from centeral servers to mobile client.

Materialized View Similar to Indexes:

- They consume storage space.
- Data update in Master table, must be refreshed.
- For query rewrite, they improve the performance
- Their existence is transparent to SQL application & user.

Materialized View Differ to Indexes:

- Materialized View can be "partitioned". / Indexes view can't be partitioned. /
- MV can be accessed directly to INSERT, UPDATE and DELETE statement.

Transaction Management

Invisible unit of work comprised of several operations, all or none which must be performed in order to preserve data integrity.

(ACID) Properties of Transaction:

1- Atomicity: The implies indivisibility; (one which will either complete fully or not at all) is atomic.

2- Consistency: Persistant data from one consistent state to another state. If a failure occures, data must be restored to the state.

3- Isolation:

Transaction should not affect each other.

like = Allied UBL — ATMs — Me

4- Durability:

Once a transaction has completed, state changes committed by the transaction must be durable and persistant.

Transaction: Commit ← End → Rollback
- 100% succefull
- 0% start from zero

↓

Series of reads and writes of database object.

↓

To read a database object, first brought data from disk to main memory. Second its value is copied into a program variable.

To write a database object, first modify and then written to disk (a copy object that's in memory).

* DB objects depend on the DBMS.

Shedule

A shedule is a list of actions from a set of transactions. (reading, writing, aborting, committing)

Interleaves:

Interleaves transaction allow to multiple users to access database at the same time.

System throughput:

The average number of transactions completed in a given time.

Serializability:

Sheduling transactions in the order in which they are executed.

Strict Two-Phase Locking (Strict 2PL):

The most widely used locking protocol, has two rules:

1- If a transaction T wants to read (respectively, modify) an object, it first requests a shared (respectively, executive) lock on the object.

2- All Locks held by a transaction are released when the transaction is completed.

Divides into 2 phase

- Growing phase
  lock aquired
- shrinking phase
  Lock released

Deadlocks:

A cycle of transactions waiting for locks to be released is called a deadlock. It's detected by wait-for graph.

The DBMS must either prevent or detect (and resolve) such deadlock situations.

Detection Vs Prevention

Lecture No 43:

→ Write Sequence

On encountering a 'write' operation, the DBMS places an entry in the log file buffer mentioning the effect of the write operation.

$X = X + 10$ before addition 23

Write X After addition 33

The entry made in the log file will be.

$<T_n, X, 33>$

- Log entry made only for the write operation.
- log file enteries have been made permanent.

Write is performed, what?

Write is performed, that is, the data from DataBase buffer is moved to disk.

→ Recovery Produce

When crash occurs, the recovery manager(RM), on restart, examines the log file from the disk. 'Checkpoint' is used in the recovery Procedure.

→ Checkpoint

Checkpoint is also a record or an entery in the log file. Check point makes the 'recovery Procedure' efficient. Logo file is maintained for 'recovery Procedure' Purpose.

Recovery Manager decides it on the following bases:

- Transactions ended before the checkpoint, are ignored.
- Transactions that have begin and an abort entry are ignored.
- Transactions that have begin and no end entry (commit or Rollback) are ignored.

∴ only one point that are Redone (Redo)

- Transitions which have both begin and the commit entries, are Redone. It does't matter if they are committed again.

MCQs:

- After execution of "commit" statement, _____ is updated first.

Database Buffer